

MUNICH STADT, Germany

WMO: 108650

Lat: 48.1632N

Lon: 11.5429E

Elev: 1724

StdP: 13.80

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.5	15.5	4.8	7.5	13.2	8.9	9.3	17.8	23.0	48.6	20.7	44.8	4.0	90	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	85.2	66.2	81.8	64.6	79.0	63.6	67.5	81.4	66.1	78.9	64.7	75.9	5.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.9	91.5	71.0	61.6	87.3	70.0	60.4	83.5	69.4	32.9	81.4	31.7	79.3	30.6	76.2	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.5	14.2	11.9	7.9	91.7	7.7	3.6	2.3	94.3	-2.2	96.4	-6.5	98.4	-12.1	101.0		
(5)			7.1	70.2	7.5	1.5	1.7	71.3	-2.7	72.1	-6.9	72.9	-12.3	74.0	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	33.0	32.8	41.0	47.3	56.4	61.7	67.1	66.1	59.6	50.6	39.9	35.2	(6)
(7)		DBStd	14.28	9.32	9.29	7.61	6.89	6.98	6.96	6.10	6.20	5.94	6.85	6.90	7.79	(7)
(8)		HDD50	2317	529	483	293	133	26	3	0	0	2	78	309	461	(8)
(9)		HDD65	6006	993	901	745	532	272	145	49	63	181	447	754	924	(9)
(10)		CDD50	2067	0	2	13	51	225	354	531	498	290	96	5	2	(10)
(11)		CDD65	282	0	0	0	0	5	47	115	96	19	0	0	0	(11)
(12)		CDH74	2131	0	0	0	3	48	334	884	726	129	7	0	0	(12)
(13)		CDH80	502	0	0	0	0	1	60	238	191	12	0	0	0	(13)
(14)	Wind	WSAvg	4.7	5.4	5.1	5.6	5.1	4.6	4.6	4.3	4.1	4.0	3.9	4.6	4.8	(14)
(15)	Precipitation	PrecAvg	38.6	2.2	1.9	2.5	2.5	4.3	5.0	5.0	4.6	3.1	2.7	2.6	2.4	(15)
(16)		PrecMax	46.7	5.6	4.2	5.2	5.5	8.1	7.8	10.6	7.8	6.0	5.2	5.8	4.5	(16)
(17)		PrecMin	27.0	0.1	0.6	0.9	0.6	0.4	2.4	1.3	1.8	0.8	0.6	0.0	0.0	(17)
(18)		PrecStd	4.6	1.2	1.0	1.3	1.4	1.8	1.8	1.9	1.7	1.4	1.2	1.3	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	60.7	69.2	71.8	78.4	85.7	91.3	89.5	81.9	74.5	60.7	57.3	(19)
(20)			MCWB	45.7	49.3	53.0	54.3	60.9	66.5	69.0	66.9	64.7	59.4	50.8	46.8	(20)
(21)		2%	DB	50.9	54.1	61.2	67.4	75.0	81.1	85.7	85.0	77.6	69.0	56.3	53.2	(21)
(22)			MCWB	43.9	45.7	48.4	52.4	59.0	63.8	66.7	65.9	62.8	57.3	48.7	45.8	(22)
(23)		5%	DB	48.1	49.1	56.6	63.6	71.8	78.0	82.1	81.4	74.3	64.3	52.9	49.4	(23)
(24)			MCWB	42.1	42.7	46.4	50.9	57.1	62.7	65.0	64.3	61.4	55.3	47.2	43.9	(24)
(25)		10%	DB	45.1	45.0	52.3	59.6	68.7	74.4	79.0	77.9	70.6	60.8	49.7	46.3	(25)
(26)			MCWB	40.5	40.4	44.6	49.0	55.8	60.8	64.2	63.1	60.1	53.9	45.7	42.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.1	49.6	53.3	55.7	63.6	67.8	70.4	68.6	66.2	61.2	52.3	49.5	(27)
(28)			MCD B	52.2	59.3	68.8	68.0	73.5	81.3	85.8	84.2	77.5	71.0	57.4	54.7	(28)
(29)		2%	WB	45.1	46.4	49.8	53.5	60.3	65.6	68.3	67.0	64.3	58.8	49.8	46.7	(29)
(30)			MCD B	50.7	53.4	58.7	65.3	70.3	77.8	82.4	81.3	74.6	65.9	54.3	51.5	(30)
(31)		5%	WB	42.7	43.3	47.4	52.0	58.5	63.8	66.5	65.6	62.7	56.4	47.9	44.6	(31)
(32)			MCD B	46.7	48.0	54.4	61.4	68.7	74.9	79.4	78.4	72.0	63.0	52.1	48.7	(32)
(33)		10%	WB	40.5	40.8	45.6	50.0	57.1	62.0	64.9	64.3	61.0	54.3	45.8	42.3	(33)
(34)			MCD B	44.4	45.0	51.9	58.3	66.5	72.3	76.1	75.1	68.9	59.7	49.5	46.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.9	11.1	12.7	14.1	15.4	14.9	15.9	16.0	14.6	12.6	10.4	9.0	(35)
(36)			MCDBR	13.6	18.6	20.3	22.1	21.1	20.9	21.8	22.3	21.1	20.2	16.5	14.3	(36)
(37)			MCWBR	8.8	11.2	10.2	10.6	9.1	7.8	8.3	7.4	8.9	10.3	10.0	9.3	(37)
(38)		5% WB	MCDBR	11.9	17.0	17.7	19.8	18.4	18.9	19.6	19.6	17.9	16.8	14.9	12.8	(38)
(39)			MCWBR	8.6	10.8	9.5	9.8	8.4	7.9	8.4	7.3	8.3	9.2	9.9	9.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.296	0.321	0.354	0.389	0.398	0.403	0.406	0.400	0.376	0.346	0.318	0.292	(40)	
(41)		taud	2.448	2.384	2.342	2.275	2.283	2.305	2.289	2.325	2.370	2.447	2.481	2.468	(41)	
(42)		Ebn at Noon	248	265	272	273	274	273	270	266	262	252	238	235	(42)	
(43)		Edh at Noon	20	27	33	39	40	40	40	37	32	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	345	625	979	1362	1571	1759	1699	1493	1081	692	374	279	(44)	
(45)		RadStd	45	83	119	180	181	158	155	151	132	81	49	36	(45)	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (91 neighbors)	+0.67	N/A	N/A	N/A	+0.59	+0.61	N/A	N/A	+99	+34	(47)

Nomenclature: See separate page